Practical-13

PHP JSON API and AJAX/Fetch-Based CRUD without Page Reload

Theory :

1. **Do PHP endpoints return valid JSON responses?**

   Answer: Yes, PHP endpoints return valid JSON responses for frontend communication.

2. **Are GET and POST requests handled properly?**

   Answer: Yes, GET and POST requests are handled correctly to retrieve and submit data.

3. **Are loading, success, and error states shown on the frontend?**
   Answer: Yes, appropriate loading indicators and success or error messages are displayed.

4. **Is input validation performed before database operations?**

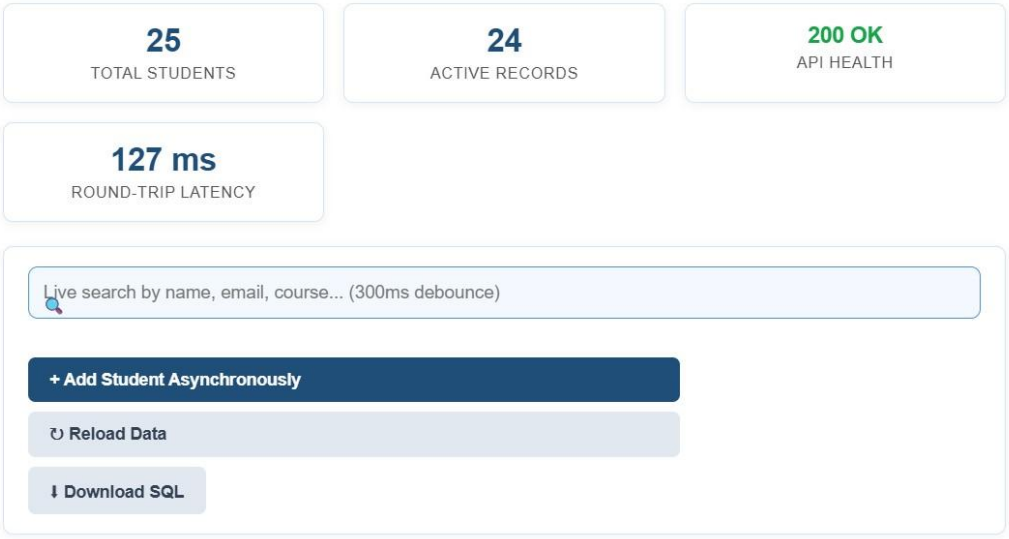
Answer: Yes, user inputs are validated before performing database operations.

Result:

1) Student panel management via api

ID	Name	Email	Mobile	Course	Year	Gender	Status
#25	Bhavya Rawal	bhavya.rawal@example.com	9711883319	BTech_AI	Year 2	Male	Inactive
#24	Varun Panchal	varun.panchal@example.com	9822668818	BCA	Year 1	Male	Active
#23	Diya Kapadia	diya.kapadia@example.com	9899114417	BTech_IT	Year 3	Female	Active
#22	Siddharth Verma	siddharth.v@example.com	9722446616	BTech_CSE	Year 2	Male	Active
#21	Drishti Patel	drishti.p@example.com	9633882215	BTech_AI	Year 4	Female	Active
#20	Parth Prajapati	parth.prajapati@example.com	9855331114	MCA	Year 2	Male	Active
#19	Krupa Vaghela	krupa.v@example.com	9744229913	BCA	Year 3	Female	Active
#18	Aditya Chudasama	aditya.c@example.com	9811993312	BTech_IT	Year 1	Male	Active

2)student panel managed by admin via api health



3)sample data management via api

#15	Ritu Rajput	ritu.rajput@example.com	9833441109	MCA	Year 1	Female	Active
#14	Yash Vora	yash.vora@example.com	9611224408	BCA	Year 2	Male	Active
#13	Tanvi Bhatt	tanvi.bhatt@example.com	9899887707	BTech_IT	Year 4	Female	Active
#12	Manav Desai	manav.desai@example.com	9744556606	BTech_CSE	Year 1	Male	Active
#11	Kavya Joshi	kavya.joshi@example.com	9855667705	BTech_AI	Year 3	Female	Active
#10	Devansh Trivedi	devansh.t@example.com	9633445504	MCA	Year 2	Male	Active
#9	Neha Mehta	neha.mehta@example.com	9711223303	BCA	Year 1	Female	Active
#8	Rohan Sharma	rohan.sharma@example.com	9822334402	BTech_IT	Year 2	Male	Active
#7	Ananya Iyer	ananya.iyer@example.com	9876543201	BTech_CSE	Year 4	Female	Active

Git Hub Repository Link:

GitHub Repository: https://github.com/justchill-spec/Student-Hub-Portal

Challenges Faced: I faced difficulties in creating PHP endpoints that return valid JSON responses. I struggled with using the Fetch API to perform CRUD operations without reloading the page. I also faced challenges in handling errors, validating inputs, and displaying appropriate messages.

Learning Outcome: I learned to create PHP JSON APIs and connect them with JavaScript using the Fetch API. I understood how to perform CRUD operations asynchronously without reloading the page. I also learned to validate user inputs and handle loading, success, and error messages.